Frontend Use Cases

Medicaid Claims Checker · test plan for the running app · status as of October 21, 2026

> **What this is.** The agreed list of things to test in the running Medicaid Claims Checker through its web pages: the Fetch Source page (`http://localhost:4000/`, Manual Upload and Batch History) and the Rules page (`/rules`). Each use case names its test files and what the app should show. Engine-only checks are out of scope.

59	59	59
use cases	with test data	checked automatically by `./test.sh`

Before you start

- The engine and Phoenix are running (`./start.sh`); the dev database has the 17 seeded active rules and the October 2026 NPPES import.
- Leave **Test upload: allow re-submission** ticked on the upload page so files can be uploaded again (see the companion guide Repeat Testing with Mock Data). Untick it only for U4–U6.
- Every path below is relative to `mock_data/`. Each use case has its own folder, `<group>/<ID>_<name>/`; the table names the folder and the files in it. The exact expected result of every generated file is in `mock_data/manifest.json`.
- Use cases with no folder of their own (N1, R8, U1–U8) use files from other folders.
- A claim "matches" a rule when the rule fires on it; "Clean" means no rule matches and the NPPES check passes.
- `./test.sh` runs all 59 through the real pages; `./test.sh R5 N7` runs only those.

J — JSON uploads

ID	Use case	Test files	What the app should show
J1	One claim in a .json file	`J_json_files/J1_one_claim/` routine_office_visit.json	"1 JSON claim(s) submitted for evaluation"; the claim is evaluated.
J2	A list of claims `[ … ]`	`J_json_files/J2_list_of_claims/` two_claims.json	Both claims evaluated: MOCK-LIST-001 Clean; MOCK-LIST-002 matches MissingDiagnosisCodes.
J3	Claims wrapped as `{"claims": [ … ]}`	`J_json_files/J3_claims_wrapper/` mixed_batch.json, clean_batch.json, mixed_risk.json	Every claim evaluated. clean_batch: 4 Clean. mixed_batch: 2 Clean, then WeekendOfficeVisit, UnauthorizedProcedure, HighComplexityERVisit, MissingDiagnosisCodes.
J4	Claims mixed with non-claim entries	`J_json_files/J4_claims_mixed_with_non_claims/` list_with_bad_entries.json	The claims are evaluated; the final message still says "Not accepted: list_with_bad_entries.json (entry …)", naming each refused entry.
J5	A JSON file that is not a claim	`J_json_files/J5_not_a_claim/` patient_record.json	Refused: "no claim found (expected a claim_id or claim field)"; nothing evaluated.

J6	An engine request (rulesText + claims)	`J_json_files/J6_engine_request/` rules_text_and_claims.json	Refused as “an engine test request (rules and a list of claims), not a claims file”; nothing evaluated.
J7	Invalid JSON	`J_json_files/J7_invalid_json/` cut_off.json	Refused: “not valid JSON”.
J8	Dental / institutional claim as JSON	`J_json_files/J8_837i_or_837d_style_json/` dental_837d.json, institutional_837i.json	Evaluated like a converted 837D / 837I; both Clean.

X — X12 (.x12) uploads

ID	Use case	Test files	What the app should show
X1	837P professional claim	`X_x12_files/X1_837p_professional/` clean_office_visit.x12, clean_multi_line.x12	“1 X12/EDI claim(s) translated”; Clean.
X2	837I institutional claim	`X_x12_files/X2_837i_institutional/` institutional.x12	Translated and evaluated; Clean.
X3	837D dental claim	`X_x12_files/X3_837d_dental/` dental.x12	Translated and evaluated; Clean.
X4	Several claims in one transaction set	`X_x12_files/X4_several_claims_one_transaction_set/` three_claims.x12, two_claims_same_subscriber.x12	Each claim is its own result row. three_claims: MULTI01 Clean; MULTI02 HighValueClaim + HighComplexityERVisit + UnauthorizedProcedure; MULTI03 MissingDiagnosisCodes. Same subscriber: SAMESUB01 Clean, SAMESUB02 SuspiciouslyLowCharge.
X5	Several transaction sets in one file	`X_x12_files/X5_several_transaction_sets/` two_transaction_sets.x12	All four claims from both sets are evaluated; TXA02 SuspiciouslyLowCharge, TXB02 PatientAgeOutOfRange.
X6	One-line file / Windows line endings	`X_x12_files/X6_one_line_or_windows_line_endings/` no_newlines.x12, crlf_line_endings.x12	Read like a normal file; Clean.
X7	Broken envelope or file	`X_x12_files/X7_broken_envelope_or_file/` truncated, wrong_se_count, control_number_mismatch, too_short, envelope_no_claims (.x12)	Refused, naming the problem: “incomplete transmission”, “SE segment count”, “interchange control number mismatch”, “too short”, “no claims found”. No claims evaluated.

E — EDI (.edi) uploads

ID	Use case	Test files	What the app should show
E1	A plain, clean .edi file	`E_edi_files/E1_plain_edi/` clean_claim.edi	Treated exactly like .x12; Clean.
E2	Alternate delimiters (`|` and `>`)	`E_edi_files/E2_alternate_delimiters/` alternate_delimiters.edi	Translated; HighComplexityERVisit + UnauthorizedProcedure.

E3	A file that is not X12 at all	`E_edi_files/E3_not_x12/` not_x12.edi	Refused as not X12 ("must start with ISA").

Z — Zip uploads

ID	Use case	Test files	What the app should show
Z1	Zip with .x12, .edi, .json and other files	`Z_zip_files/Z1_mixed_file_types/` mixed_bundle.zip	The three claim files are evaluated (ZIPEDIB matches MissingDiagnosisCodes); README.txt is ignored.
Z2	Zip with JSON files only	`Z_zip_files/Z2_json_only_zip/` two_json_claims.zip	Both JSON claims evaluated; Clean.
Z3	Path-traversal entry (`../escape.x12`)	`Z_zip_files/Z3_path_traversal/` path_traversal.zip	Read as escape.x12 and evaluated (Clean); nothing is written to disk.
Z4	Zip with nothing usable / empty zip	`Z_zip_files/Z4_nothing_usable/` empty.zip, text_files_only.zip	"No processable claims found".
Z5	Zip inside a zip	`Z_zip_files/Z5_zip_inside_zip/` nested.zip	The outer .x12 is evaluated (Clean); the inner zip is ignored.
Z6	Corrupt zip	`Z_zip_files/Z6_corrupt_zip/` cut_off.zip	"No processable claims found".
Z7	More than 15,000 claims	`Z_zip_files/Z7_over_15000_claims/` — not in git; make it with `elixir mock_data/generate.exs --large 15001`	Nothing is evaluated. The message names the reason: "This batch has 15001 claims; the limit is 15000. Split the files into smaller batches and try again." Up to 15,000 claims are evaluated normally.

U — Upload behaviour

ID	Use case	Test files	What the app should show
U1	Several files of mixed types at once	For example R2_each_rule_alone/HighValueClaim.json, X1…/clean_office_visit.x12, Z1…/mixed_bundle.zip	Every file's claims are evaluated in one batch.
U2	Unsupported file type (.txt, .pdf)	Any non-claim file	"File type not accepted. Allowed: .json, .x12, .edi, .zip", before upload.
U3	File over 10 MB	Any file over 10 MB (not kept in git)	"File is too large (max 10 MB).", before upload.
U4	Same file uploaded twice	Any claim file, checkbox unticked	Second time: "Nothing was evaluated: every claim was already submitted…", and a red notice in Batch History.

U5	Same claim in a different file	A copy of an uploaded file with one extra line at the end, checkbox unticked	The claim is refused as already submitted, naming the earlier batch; other claims are evaluated.
U6	Same claim twice in one file	A JSON list holding one claim twice, checkbox unticked	Only the first copy is evaluated; the second is refused (“appears twice in this submission”).
U7	Void / replacement (frequency 8 / 7)	`R_business_rules/R11_replacement_or_void_claim/`	Accepted: a void or replacement is not a duplicate of the original.
U8	Re-uploading test data	Any file, with **Test upload: allow re-submission** ticked	The same file is evaluated again; the batch name ends in “(test re-submission)”.

R — Business rules (17 active rules)

ID	Use case	Test files	What the app should show
R1	Clean claim	`R_business_rules/R1_clean_claim/` office_visit, outpatient_lab, preventive_care, specialist_consultation (.json)	No rule matches; Clean.
R2	Each rule on its own	`R_business_rules/R2_each_rule_alone/` <RuleName>.json and .x12, one per rule	The named rule matches. Rules that always go together match too (ExtremeAmounts brings ExcessiveTotalCharges and HighValueClaim); the X12 files also match UnauthorizedProcedure where the claim has no authorization.
R3	Several rules on one claim	`R_business_rules/R3_several_rules/` 4 JSON files, many_rules.x12	Every expected rule matches (manifest.json lists them).
R4	Problem on the 2nd service line	`R_business_rules/R4_problem_on_second_line/` er_99285_line2, future_date_line2, pre1900_line2 (.x12)	HighComplexityERVisit, FutureServiceDate and ImpossibleDates match: rules check every line, not just the first.
R5	Exact thresholds	`R_business_rules/R5_exact_thresholds/` exactly_5, exactly_500_weekend_office, exactly_5000, exactly_50000 (.x12)	“Greater than” limits don’t match at exactly the limit (exactly_50000 matches only UnauthorizedProcedure).
R6	Weekend office visit variants	`R_business_rules/R6_weekend_office_variants/` saturday_office, sunday_office, weekday_office_no_match, saturday_home_visit_no_match (.x12)	Saturday and Sunday office visits match WeekendOfficeVisit; weekday and home visits don’t.
R7	Weekend inpatient stay	`R_business_rules/R7_weekend_inpatient_stay/` inpatient_weekend.x12, inpatient_weekend_stay.json	WeekendOfficeVisit doesn’t match; Clean.
R8	New-provider rules (tenure)	`R_business_rules/R2_each_rule_alone/` NewProviderLargeSubmission.json, TaxonomyMismatchNewProvider.json	Match on JSON claims (tenure comes only from JSON).

R9	Rule that cannot be decided	`R_business_rules/R9_rule_not_decidable/` missing_taxonomy_new_provider.x12	TaxonomyMismatchNewProvider shown as “Not checked”, naming the missing field.
R10	Patient age out of range / missing DOB	`R_business_rules/R10_age_out_of_range_or_missing_dob/` born_1880, born_2099, missing_dob (.x12)	PatientAgeOutOfRange for 1880 and 2099; MissingRequiredFields for no date of birth.
R11	Void / replacement against business rules	`R_business_rules/R11_replacement_or_void_claim/` replacement_frequency_7, void_frequency_8 (.json and .x12)	The rules apply as to an original claim: HighValueClaim matches (the X12 files also match UnauthorizedProcedure).

N — NPPES provider checks

ID	Use case	Test files	What the app should show
N1	Active provider	Any clean claim, e.g. R1_clean_claim/	NPPES check passes.
N2	Malformed NPI (length / check digit)	`N_nppes_provider_checks/N2_malformed_npi/` nine_digit_npi.x12, bad_checksum.x12	InvalidProviderNPI (and InvalidNPIFormat for 9 digits); NPPES: “not found in NPPES”.
N3	NPI never issued	`N_nppes_provider_checks/N3_npi_never_issued/` unissued_npi (.json and .x12)	“not found in NPPES”.
N4	Deactivated provider	`N_nppes_provider_checks/N4_deactivated_provider/` deactivated_provider (.json and .x12)	“was deactivated”.
N5	Service before the deactivation	`N_nppes_provider_checks/N5_service_before_deactivation/` service_before_deactivation.x12	Accepted; Clean.
N6	Deactivated part-way through a claim	`N_nppes_provider_checks/N6_deactivated_mid_claim/` deactivated_mid_claim.x12	Rejected for the later line: “service rendered on 2024-12-20 …”.
N7	Reactivated provider / service in the gap	`N_nppes_provider_checks/N7_reactivated_provider/` reactivated_provider, reactivation_gap (.json and .x12)	Accepted after reactivation; “was deactivated” for service inside the gap.
N8	Service before the NPI was issued	`N_nppes_provider_checks/N8_service_before_npi_issued/` before_enumeration.json	“was not issued until …”.
N9	Provider dropped from the NPPES file	`N_nppes_provider_checks/N9_dropped_from_nppes/` dropped_provider.json	“has not been in the NPPES file since …” (needs the October 2026 import).
N10	No billing NPI	`N_nppes_provider_checks/N10_no_billing_npi/` missing_billing_npi.x12	MissingRequiredFields.
N11	Deactivated rendering provider	`N_nppes_provider_checks/N11_deactivated_rendering_provider/` rendering_npi_deactivated.x12	NPPES rejects the rendering provider: “was deactivated”.

P — Rules page

ID	Use case	Test files	What the app should show
P1	Feature examples	`P_rules_page/P1_feature_examples/` examples.dsl	Each of the 10 rules parses, passes the schema check and saves as an active rule.
P2	Redundancy levels	`P_rules_page/P2_redundancy_levels/` redundancy_levels.dsl (add in file order)	Exact duplicate, subsumption, shadowed and overlap warnings each appear where the file's comments say.
P3	Rule with a syntax error	`P_rules_page/P3_syntax_error/` missing_then.dsl	"Rule syntax issue" shown while typing; saving gives "Could not save rule"; nothing saved.
P4	Rule naming an unknown field	`P_rules_page/P4_unknown_field/` old_field_name.dsl	Saved as an inactive draft, with the current name suggested ("subscriber.subscriber_id -> subscriber.member_id").